



A Prescriptive Analytics Driven Decision Support System for Pharmaceutical Demand Forecasting and Planning in MedShield Pharmaceutical Corporation

**A Capstone Project Progress Report Presented to the
Department of Information Systems
College of Information and Computing Sciences
University of Santo Tomas**

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of the requirements for the degree of
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Chapter 4

Results and Discussion

This progress report presents the updated implementation and verification results of the MedShield Pharma Corporation Decision Support System. The current revision advances the dashboard from a static prototype toward a source-backed analytical application. It introduces calendar-aware views, revised product and sales visualizations, explicit buyer clusters, rolling forecast horizons, and scenario-safe prescriptive planning. The system is implemented and testable, but selected data-quality and operational-approval gates remain open.

4.1 Presentation of Results

The revised DSS now operates through a Next.js dashboard, a TypeScript API gateway, and Python analytical services. The application supports multi-year data from 2017 onward, dynamic calendar behavior, transaction-based buyer classification, descriptive product and territory analysis, baseline forecast validation, and constrained planning scenarios. The changes are recorded in Git commit 4d00c29 on branch `ega_medshieldup3`.

The implementation distinguishes descriptive evidence, forecast estimates, and scenario outputs. Actual observations override estimates whenever current-year actual data becomes available. Missing observations remain unavailable rather than being converted to zero. Forecast and prescriptive outputs remain draft or scenario evidence and are not treated as approved procurement instructions.

Component	Implemented progress	Current qualification
Calendar coverage	All Years spans 2017 through the current calendar year and advances automatically.	2017-2019 Net CP revenue remains unresolved.
Overview and Sales Diagnostics	Single Year and Y/Y use monthly granularity; All Years uses annual granularity.	Current-year estimates stop at the current Philippine month.
Product Prioritization	Revenue Pareto with cumulative contribution, ABC, portfolio, and scoped trends.	Y/Y is removed; the companion trend is monthly or yearly.
Buyer clustering	Government, Private, Internal, and Unknown are shared across DSS modules.	Classification reflects defined source-label rules.
Forecasting	Current-month windows support 3, 6, and 12 months with historical validation.	Baseline models remain draft and source age is disclosed.
Prescriptive planning	Pareto shortlist and integer-pack allocation scenario.	Operational inputs still require approval.



4.2 Data Analysis

The updated analytical design follows a consistent time-grain rule. A selected year is examined by month because month-level movement is more useful for identifying seasonality, changes in sales activity, and the timing of planning decisions. All Years is summarized annually to preserve long-term comparability without crowding the interface. Y/Y comparison is retained only where direct month-aligned comparison is analytically meaningful.

Product prioritization now uses the Pareto principle directly. Products are ranked from highest to lowest net sales revenue, shown as bars, while cumulative contribution is plotted as a percentage line. ABC classes, the product table, portfolio view, and companion time trend are recalculated from the same selected period.

Buyer classification is expressed through four mutually exclusive clusters. Government includes national government, public hospitals, and accounts explicitly named as LGUs or government institutions. Private includes generic provincial sales areas not explicitly named as LGUs, hospital and pharmacy accounts, and individual sales accounts. Internal isolates MedShield business labels. Remaining records are Unknown.

Buyer cluster	Observed records	Interpretation
Government	7,544	Explicit national government, public hospital, or LGU labels.
Private	22,523	Generic provincial, hospital, pharmacy, or individual-account labels.
Internal	4,901	MedShield administration and internal business-line records.
Unknown	577	Records not classifiable under the other rules.

These are classification coverage counts from the September 15, 2026 live verification, not market-share estimates. Revenue and quantity shares are calculated only within the selected cluster and analytical scope.

4.2.1 Financial Data Limitation

Local records for 2017, 2018, and 2019 currently contain zero values in the approved Net CP field used for net sales revenue. Gross sales fields exist, but substituting them would change the approved financial definition. Historical ingestion and year-specific financial mappings must be repaired and reconciled before those years can be presented as validated revenue.



4.3 Dashboard

The dashboard integrates descriptive, predictive, and prescriptive views into a common interface. Time filters change the underlying aggregation, buyer clusters use one shared rule, actual observations remain distinct from estimates, and scenario recommendations display their evidence and limitations.

4.3.1 Executive Overview

The Executive Overview summarizes cumulative revenue, current-year outlook, peak periods, and leading customer context. A selected year displays January through December, Y/Y comparison aligns both selected years by calendar month, and All Years displays annual totals from 2017 through the current year.

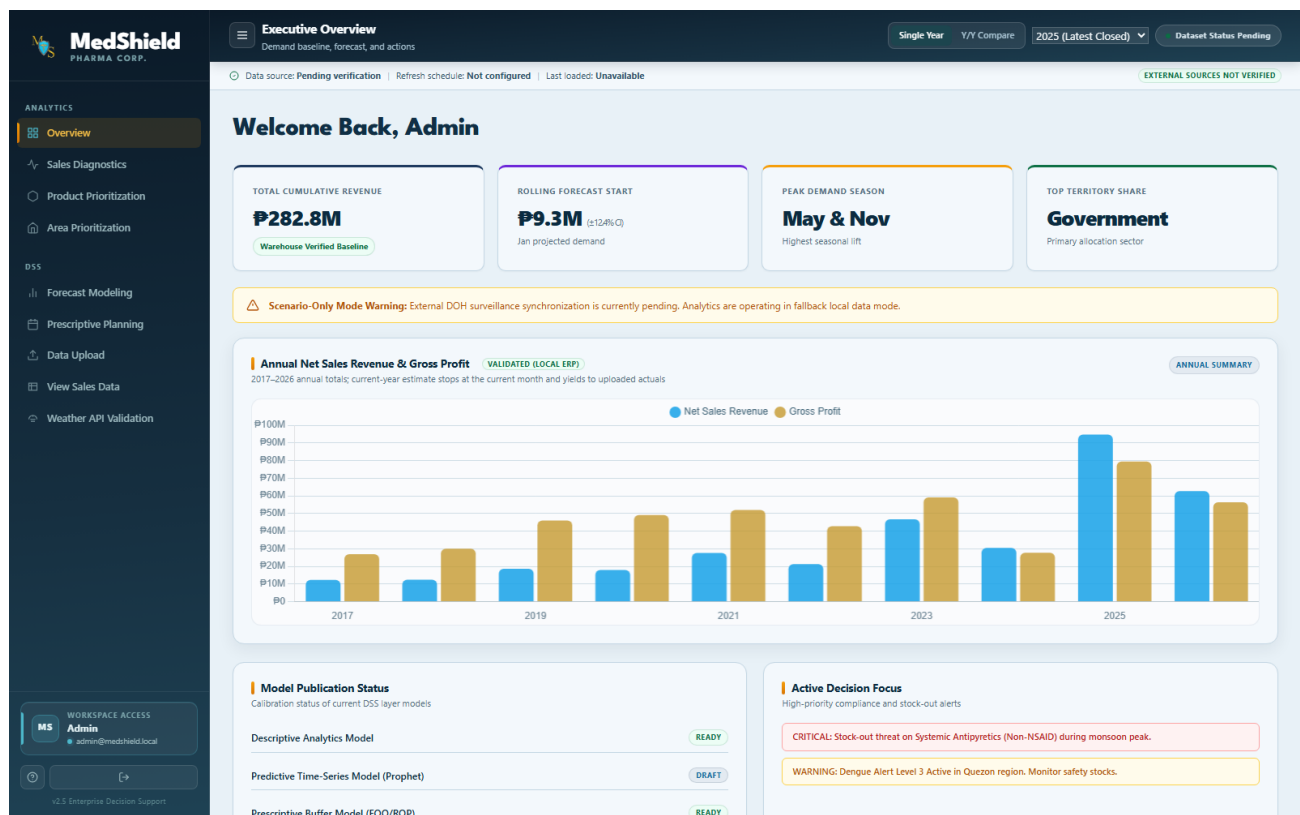


Figure 4.1 Updated MedShield DSS Executive Overview

The 2026 annual value is labeled as an estimate through the current month. Philippine calendar time determines the cutoff. When actual monthly data is loaded, it replaces the corresponding estimate while later unavailable months remain outside the displayed current-year evidence.



4.3.2 Sales Diagnostics

Sales Diagnostics applies the same time-grain rules to net sales, gross profit, growth, and margin analysis. Single Year displays monthly values, Y/Y compares matching months, and All Years retains an annual comparison. Percentage growth is accompanied by nominal peso movement, and missing months are not treated as zero sales.

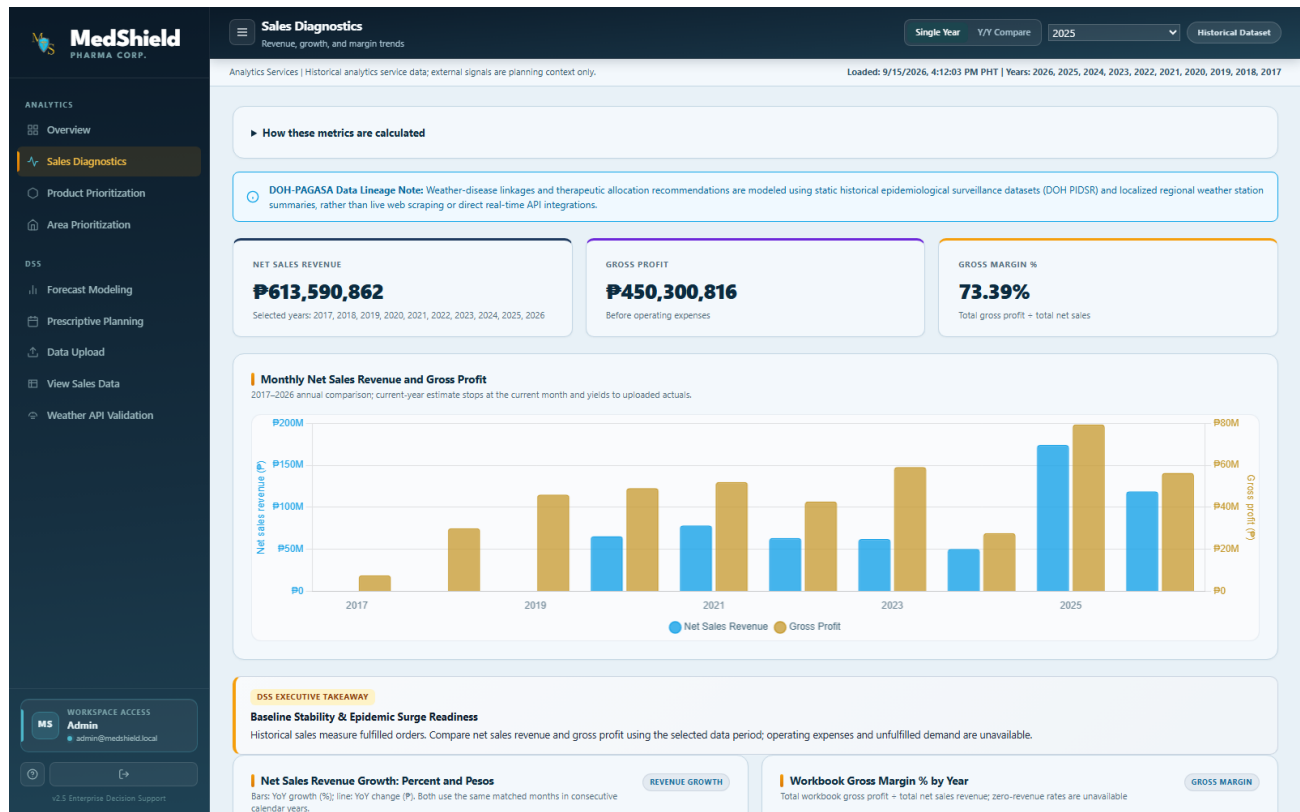


Figure 4.2 Updated Sales Diagnostics with Dynamic Time Grain

Gross margin is derived from aggregate gross profit divided by aggregate net sales for the selected scope. Labels distinguish revenue, recorded gross profit, rates, and delivered source units to reduce metric ambiguity.



4.3.3 Product Prioritization

Product Prioritization no longer offers Y/Y comparison because it ranks the portfolio rather than comparing two annual series. Its primary visualization is a Pareto chart: descending net sales revenue bars and cumulative revenue contribution on a secondary percentage axis.

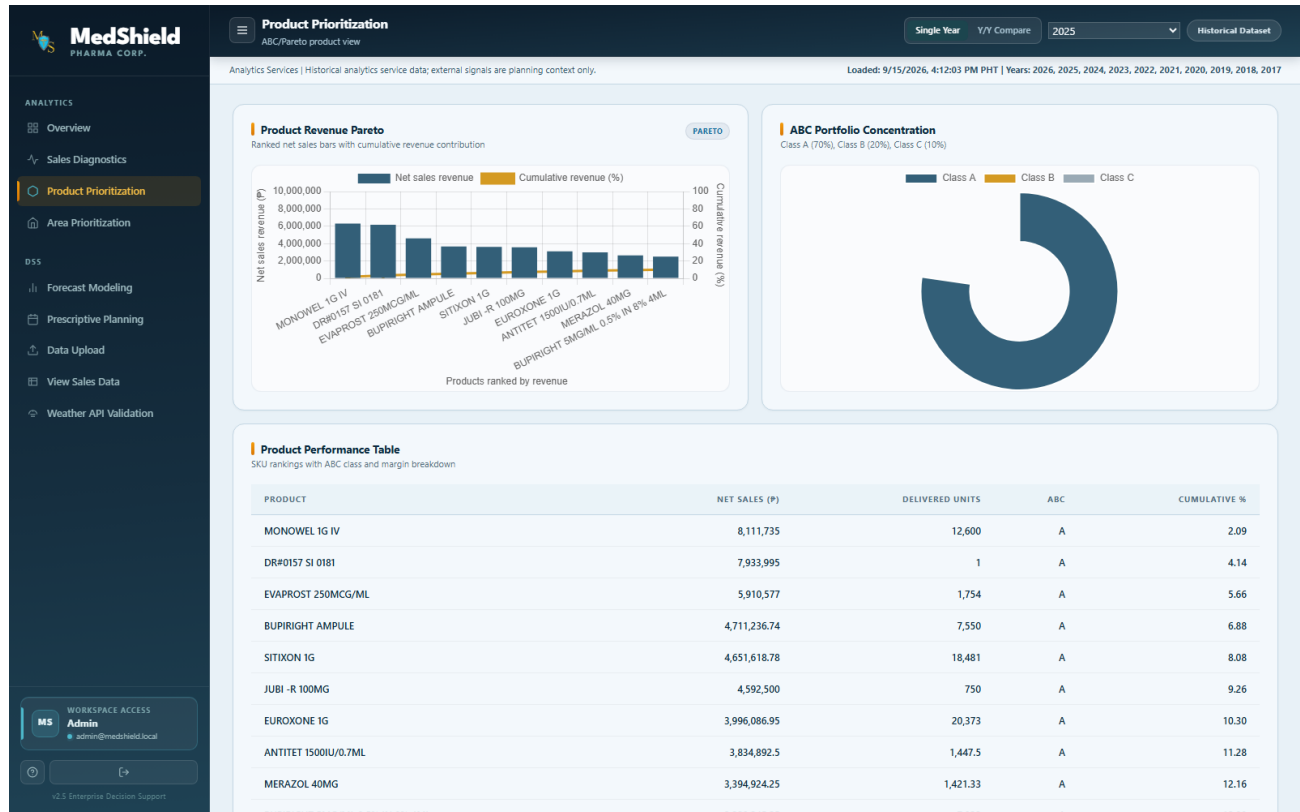


Figure 4.3 Product Revenue Pareto and ABC Portfolio Views

For a selected year, the companion trend shows monthly revenue for leading products. Under All Years it changes to yearly granularity. ABC distribution, portfolio position, cumulative contribution, and ranking rows are recalculated from the same selected period.



4.3.4 Area Prioritization

Area Prioritization separates buyer ownership from geography and customer channel. Users select Government, Private, Internal, or Unknown, then examine revenue share and product-specific quantity share by geography or channel.

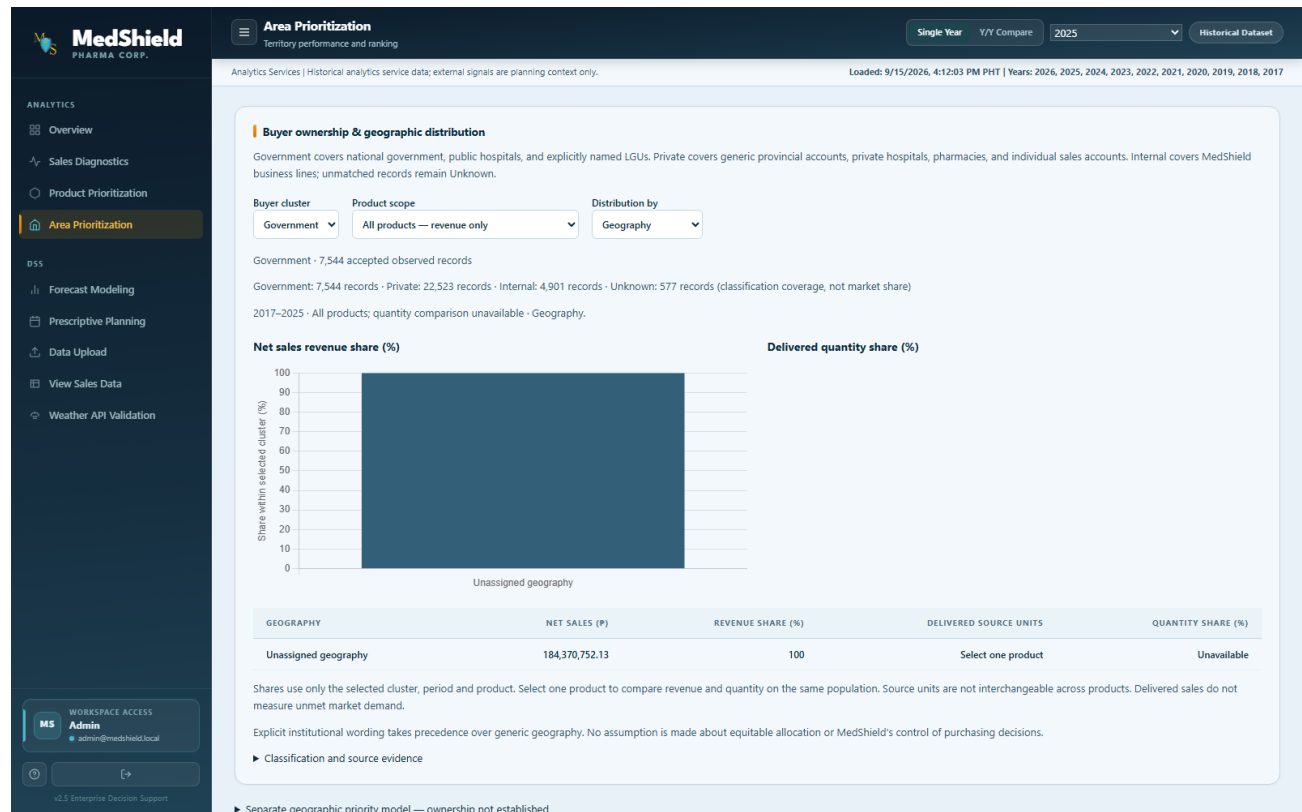


Figure 4.4 Buyer Cluster and Geographic Distribution Analysis

Explicit institutional wording takes precedence. A province name alone is treated as a private sales area, while an explicitly named LGU or provincial government is Government. MedShield business-line labels are Internal. Unknown remains visible so records are not silently reassigned.



4.3.5 Forecast Modeling

Forecast Modeling begins with the current Philippine calendar month. The selector provides 3, 6, and 12 months, with the current month counted as month one. On September 15, 2026, the windows are September-November 2026, September 2026-February 2027, and September 2026-August 2027.

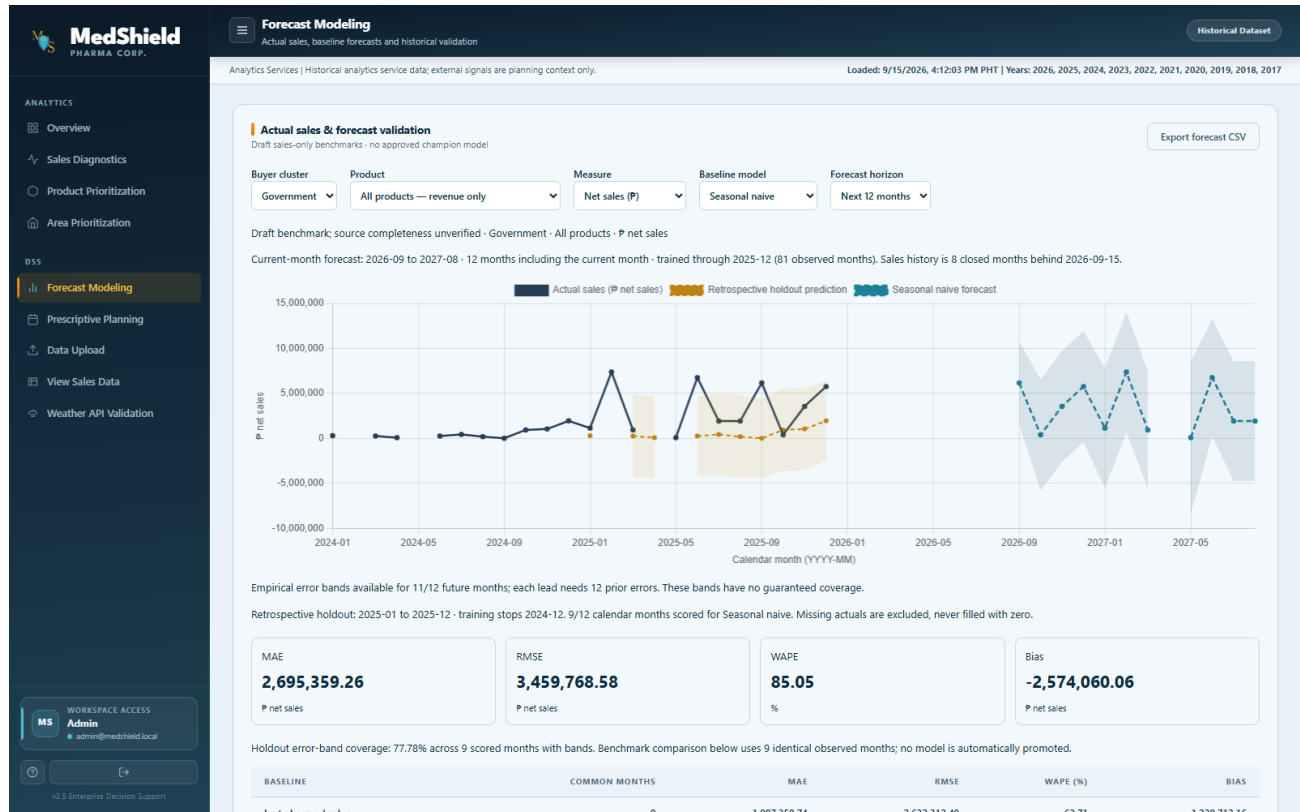


Figure 4.5 Current Month Forecast and Historical Validation

The chart retains actual observations, retrospective holdout predictions, future baseline forecasts, and empirical historical-error ranges. Missing calendar months remain visible as gaps. The service discloses the training origin and age of available sales history. Seasonal naive and last-observed-value remain draft benchmarks; MAE, RMSE, WAPE, bias, coverage, and origin support evaluation but do not establish an approved champion.



4.3.6 Prescriptive Planning

Prescriptive Planning uses the same buyer clusters. It ranks products by positive net sales over the latest available 12 months, selects the top 20 percent capped at five products, and prepares an integer-pack allocation scenario.

MedShield PHARMA CORP.

Prescriptive Planning
Priority products, constrained allocation and scenario review

Analytics Services | Historical analytics service data; external signals are planning context only. Loaded: 9/15/2026, 4:12:03 PM PHT | Years: 2026, 2025, 2024, 2023, 2022, 2021, 2020, 2019, 2018, 2017

Priority-product allocation scenario
Maximize average fulfillment across shortlisted products, then minimize purchase cost. This objective requires client review. Results are planning scenarios, not purchase orders.

Buyer cluster: Unknown Territory: Quezon Planning horizon: 3 months Purchase budget (P): Minimum fulfillment per product (%): 0

No eligible positive-revenue products in this scope. Private ownership requires approved buyer mappings.

The shortlist is the top 20% of positive-revenue raw product identities, capped at five, ranked over the latest 12 calendar months in this scope. Its actual revenue share is shown; 80% coverage is not assumed.
Enter total demand for the selected horizon, usable stock, protected stock, pack size in source units, cost per pack, and supplier pack limit. Blank inputs are unknown. Historical quantity is reference evidence—not a forecast. Changing the horizon requires reviewing total demand.

PRODUCT / HISTORICAL REFERENCE	HORIZON DEMAND	USABLE STOCK	PROTECTED STOCK	UNITS PER PACK	COST PER PACK (P)	SUPPLIER LIMIT (PACKS)
☐ I have reviewed demand for this horizon and confirm these are scenario assumptions, not verified inventory or procurement data.						

Solve allocation scenario **Export scenario CSV**

WORKSPACE ACCESS
MS Admin admin@medshield.local
v2.5 Enterprise Decision Support

Figure 4.6 Buyer Scoped Prescriptive Planning Scenario

The objective prioritizes fulfilled-demand fractions and then minimizes cost at the best attainable service level. Demand, usable and protected stock, pack size, price, supplier limits, budget, and minimum fulfillment are required. Results remain scenario-only until operational inputs and policies are approved.



4.3.7 Data Upload Page

The Data Upload page retains source-import controls and an ingestion status log. It supports continued loading of sales and structured data without hard-coding the final year. Current-year actuals are designed to supersede estimates when new monthly records arrive.

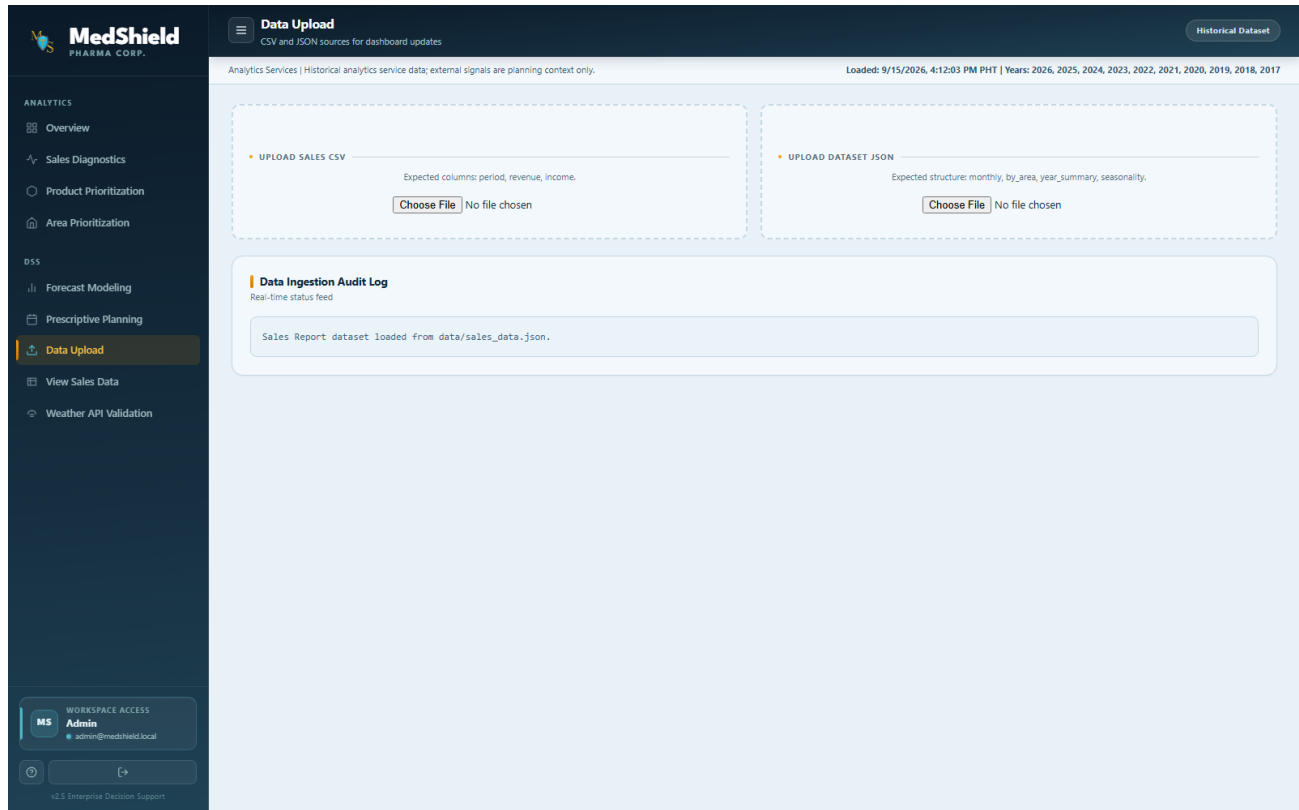


Figure 4.7 Updated Data Upload and Ingestion Status Page

The workflow remains subject to validation, duplicate handling, financial reconciliation, and lineage. The unresolved 2017-2019 Net CP issue must be corrected before a refreshed snapshot can be described as financially validated.



4.3.8 MedShield DSS KPI Tree

The KPI tree connects the system objective to the measures required for sales monitoring, product and territory prioritization, forecast evaluation, planning effectiveness, and data governance. Data trust governs every branch, while forecast quality and product priority feed the planning scenario.

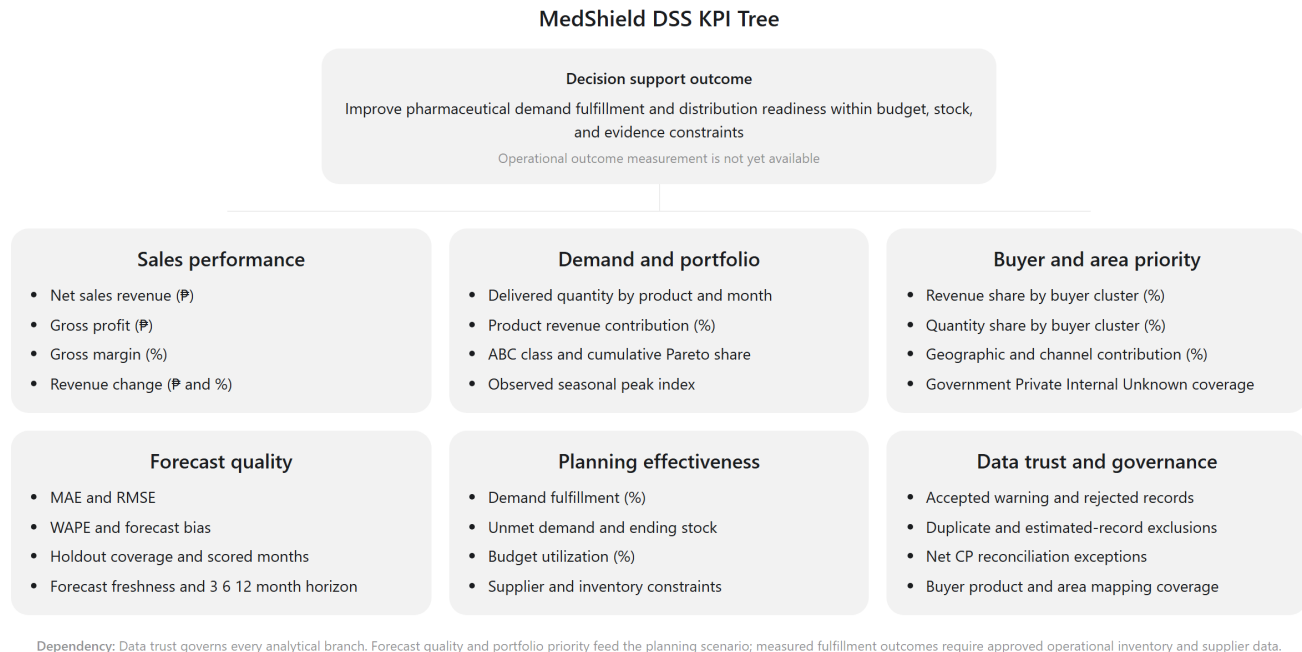


Figure 4.8 MedShield DSS KPI Tree

The top-level operational outcome is not yet measured because actual inventory, supplier, and fulfillment data are incomplete. The current DSS therefore reports analytical and scenario KPIs while preserving the distinction between observed performance, forecast diagnostics, and planned allocations.



4.4 Summary of Findings

The current progress establishes a functioning analytical dashboard with source-aware controls rather than a static design demonstration. Major improvements are consistent monthly and annual time grains, the product Pareto view, current-year estimates replaceable by actuals, the four-cluster buyer model, current-month forecast windows, and alignment of prescriptive planning with the same cluster definitions.

Limitations remain material. The 2017-2019 revenue fields are unresolved, current forecasts use sales history ending in December 2025, product identities and units are not fully standardized, and prescriptive outputs lack approved operational inputs. These constraints prevent operational certification but do not negate implementation and verification of the DSS workflow.

Finding	Evidence	Status
Dynamic calendar behavior	2017-current All Years range and Philippine-month cutoff.	Implemented
Product analytical method	Revenue Pareto, cumulative line, ABC, and scoped trend.	Implemented
Buyer segmentation	Four clusters shared across three DSS modules.	Implemented
Forecast horizon	Current month plus selectable 3, 6, or 12 months.	Draft benchmark
Prescriptive support	Pareto shortlist and constrained integer-pack allocation.	Scenario only
Historical revenue	Net CP is zero for 2017-2019 in the current source.	Blocked



4.5 Refinement of Analytical Outputs Based on Technical Consultation

The latest revision implements the principal recommendations documented during the September 2026 consultation. Revenue and profit comparisons now use clearer time grains; growth retains nominal and percentage evidence; product prioritization uses a direct Pareto representation; buyer behavior is separated into explicit clusters; forecast views include actual and retrospective evidence; and prescriptive output is constrained to a focused shortlist.

The product module was refined further by removing Y/Y comparison, converting the primary chart to a true revenue Pareto, and retaining monthly or annual movement in a companion trend. This separates portfolio concentration from time-series behavior while keeping both under the same selected period.

Forecasting changed from a static future year to a rolling window beginning in the current month. Closed-month observations remain the training population, and stale-history warnings remain visible. Sector segmentation now follows one precedence rule across area, forecast, and planning modules.

4.5.1 Verification Record

Verification activity	Observed result
Backend analytical regression suite	29 tests passed across clustering, forecast, regression, and planning.
Frontend type validation	TypeScript no-emit validation passed.
Targeted browser verification	Four cluster and forecast cases plus the dedicated Pareto case passed.
Live service check	Forecast start returned 2026-09 and cluster counts reconciled.
Version control	Commit 4d00c29 pushed to branch ega_medshieldup3.

4.5.2 Remaining Work

The next priority is to repair and reconcile year-specific financial extraction for 2017-2019 without substituting gross sales for Net CP. The dashboard snapshot and downstream analytical populations should then be regenerated from one approved source. Product aliases and units, current inventory, supplier constraints, and authenticated acceptance evidence should be completed before operational claims are made.